Heat-engine efficiency

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Learning target: Calculate useful output and rejected energy for a heat engine.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 6.*
Same average per particle; total translational energy is 3 times as large.

02. Worked example: explain the model
100=25+75 J; useful-work fraction 25%.

03. Read and interpret the model
Balances; 30% efficiency.

04. Apply the physics
25%; 150 J.

05. Explain your reasoning
Motor converts electrical work, not a cyclic heat-engine input; different constraints and input energy forms.

06. Review: connect an earlier idea *Review: Unit 4, Day 19.*
Both become 1/16 of their original values; both scale with v^2.

07. Exit check
40%; 60 J.

08. STRETCH: test the model
10 J unaccounted; could be stored energy or another transfer, or inconsistent measurements.

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01. Warm-up: bring back an earlier idea *Review: Unit 5, Day 6.*
Rms speed becomes 3 times as large because mean kinetic energy is proportional to the square of rms speed.

02. Worked example: explain the model
100=25+75 J; useful-work fraction 25%.

03. Read and interpret the model
Balances; 30% efficiency.

04. Apply the physics
Max 50%, max 400 J; actual 30%.

05. Explain your reasoning
Motor converts electrical work, not a cyclic heat-engine input; different constraints and input energy forms.

06. Review: connect an earlier idea *Review: Unit 4, Day 19.*
Coupled energy becomes 4/6≈0.6667 of the original; multiply the mass and fraction factors.

07. Exit check
450 J; 30%.

08. STRETCH: test the model
Bound 50%; actual 40%, below the bound. Rejected heat 600 J if steady cycle.